

Launch Readiness — The First 1,000 Clients

50pick · 50pick.tz · 3 September 2026 · production commit 78c3876e . All figures measured on the live system today.

1 · What we pay now

Service	Per month	What it is for
Railway — 50pick hosting	\$23.57	The application, database and cache
Railway — five other projects	\$8.24	Amanat Watan, Intercontinental Glass, LMC, two others
Anthropic — AI	\$50.75	Writing markets and auto-settling results
Cloudflare R2	~\$1.00	Customer ID documents and backups
Postmark email	\$0.00	Inside the free allowance — 20 emails sent
Sentry	\$0.00	Free tier
SMS	\$0.00	Not sending anything — see item 1 below
Twelve Data price feed	not measured	Billed outside Railway — needs confirming
Total today	≈ \$84	

Railway's last four bills were \$5.00, \$11.15, **\$25.42** and \$17.38 so far this period. The cost is rising because the platform is doing more work, not because anything is wrong.

Two facts that decide where the money goes.

The AI bill is bigger than the hosting bill — \$50.75 against \$31.81 — and **99.98% of it does not depend on customer numbers**. It is the cost of creating and settling markets on a clock. Going from 104 customers to 1,000 will barely change it.

Memory is 87% of the 50pick hosting bill — \$20.60 of \$23.57. The application uses almost no processing power and holds 4.7 GB of memory. Capping that is the cheapest saving we have.

2 · What we will pay with 1,000 clients

Sized for 1,000 registered customers with 100–200 online at once.

Service	Now	At 1,000	Reason it changes
Railway — 50pick	\$23.57	\$56 – 107	More visitors, more memory, more traffic
Railway — other projects	\$8.24	\$8 – 10	Unchanged
Anthropic — AI	\$50.75	\$55 – 110	Market creation is flat. Only customer chat grows.
SMS sign-up codes	\$0	\$35 – 45	New cost — about 4,000 messages a month
Postmark email	\$0	\$0 – 15	Free to 100 messages a month
Cloudflare R2	~\$1	\$1 – 3	More ID documents stored

Total	≈ \$84	\$155 – 290	
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Budget \$200 per month for normal running. Section 4 covers what a bad month costs and which lines have no ceiling today.

3 · What must be done before we open

	Item	Cost	Description
●	Turn on SMS	\$35–45/mo	No member of the public can register today. Sign-up requires a code by text message; the platform writes those codes to a log file instead of sending them, and has never sent one. Requires a signed Selcom contract and a TCRA sender ID.
●	Two-factor on the admin console	\$0	Nine administrator accounts control customer money and are protected by a password only.
●	Health check on deployments	\$0	Railway does not check whether a new version works before sending customers to it. The status page also reports healthy while the database is down.
●	Continuous database backup	\$2–5/mo	Backup runs once a night. A failure at the wrong hour loses up to 24 hours of bets, deposits and payouts. Point-in-time recovery is included in the current plan and is not switched on.
●	Automatic failure alerts	\$0	In August the nightly backup failed eleven nights running. The warning displayed correctly for ten of them and was not seen. Nothing currently sends an alert.
●	Updates without downtime	\$0	Every release briefly takes the site down and cancels any bet in progress.
●	Payouts on large markets	\$0	Winners are paid one at a time inside a thirty-second limit, which runs out at about 660 winners. Beyond that the market freezes for betting and cashing out. Largest market to date: 29 bets.
●	Cache the busy screens	\$2–4/mo	Nothing is cached. Every page is rebuilt from the database on every view and the main screens refresh every 15–20 seconds — about 400 database queries a second at 200 customers online.
●	Move servers to Amsterdam	\$0	Servers are in California, customers in Tanzania. Server responds in 31 ms; customer waits about 510 ms. Moving halves the wait. Requires one short planned outage.
●	Enlarge the database disk	\$0 now, \$7.50/mo when full	897 MB of 5 GB, growing 14 MB a day — about six months remaining. The compliance record

			must be kept seven years and cannot be deleted. Billed on space used, not reserved.
●	Recovery procedure and rehearsal	\$0	No written restore procedure and no measured recovery time. The restore tool was rehearsed once, in July, against a database a tenth of today's size.
	Added monthly cost	\$39–54	Already included in the \$155–290 total above

4 · Emergency and contingency costs

Three scenarios. Normal is the planned figure. Mid is a heavy month. Max is several problems in the same month.

Line	Normal	Mid	Max	What drives it
Railway — 50pick	\$56–107	\$137	\$515	Emergency scaling to 4 copies; traffic under attack
Railway — other projects	\$8–10	\$10	\$10	Unchanged
Anthropic — AI	\$55–110	\$150	\$700	Spend limit is \$100 per cycle but the limit fails open on error
SMS	\$35–45	\$120	\$260	Registration surge; sign-up abuse before it is blocked
DDoS protection	\$0	\$20	\$200	Cloudflare Pro, or Business during an attack
Email	\$0–15	\$15	\$50	Volume above the free allowance
Error reporting	\$0	\$26	\$26	Paid tier once error volume rises
Cloudflare R2	\$1–3	\$3	\$5	ID documents stored
Total per month	\$155–290	\$481	\$1,766	

What is already limited, and what is not

Risk	Limit in place today	Exposure
AI overspend	\$100 per cycle. Currently \$71.48 spent of \$100.	The limit fails open. If the meter itself errors, spending continues unmetered. This is the largest uncontrolled line.
SMS pumping	5 codes then 1 per 2 minutes per phone; 3 sign-ups per hour per phone; 10 new phones per address per 20 minutes.	Bounded. A determined attacker across ten addresses reaches roughly 10,000 messages a day — about \$100/day — and is visible immediately.
Traffic under attack	Railway caps 10,000 requests per second per host.	Unlimited cost. Traffic is billed per gigabyte and there is no firewall in front of the site. This is the only line with no ceiling.
Database disk	Billed on space used; can grow to 1 TB.	\$0.15 per GB per month. A full disk stops the platform accepting bets until resized.
Runaway scaling	None. Railway bills what is consumed.	Set a spending limit on the Railway account. It is not set today.

Three spending limits worth setting before launch, all free: a hard spend cap on the Railway account; an alert when the AI cycle passes 80% of \$100; and Cloudflare in front of the site, which removes the only line that has no ceiling. Together these cap the Max scenario at roughly **\$600** instead of \$1,766.

5 · Decisions taken

Question	Decision
How many customers online at once?	Size for 100–200 at peak
Selcom SMS contract	In progress — staged as a reversible switch, ready the day it lands
Server location	Move to Amsterdam in an announced maintenance window
Run several copies of the application?	No. One container serves 200 online comfortably once caching is in. Running several would first require rebuilding emergency controls that currently reach only one copy.

Part Two · Technical detail

MEASURED 3 SEPTEMBER 2026

Architecture as deployed

Players (Tanzania)
| no CDN, no firewall, every response uncached
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Railway edge network ----> ORIGIN: us-west2 (California) -- all three services
|
+-- 50pick Next.js 15, RAILPACK 1 container
| no health check, no deploy overlap, no graceful shutdown
+-- PostgreSQL 18.6 1 container, 5 GB disk (897 MB used)
+-- Redis 1 container, connected, cross-container

External services
Selcom deposits and withdrawals LIVE (apigw.selcommobile.com/v1)
Postmark transactional email LIVE, 20 sent, 0 failures
Cloudflare R2 . KYC documents + backups LIVE
Sentry error reporting LIVE, PII scrubbing proven
Anthropic market generation + settling ... LIVE
Twelve Data ... Up & Down price oracle LIVE
SMS one-time codes NOT DELIVERING (provider = console)

31 ms
server p50 (p95 136 ms)

0
5xx errors in 500 requests

4.68 GB
average memory (peak 8.86)

14 MB
disk growth per day

Measured platform state

Item	Measured value
Railway plan	Pro — 42 replicas allowed, volumes to 1 TB, 10,000 connections per host
Deploy configuration	numReplicas 1, healthcheckPath null, overlapSeconds null, drainingSeconds null on all three services
PostgreSQL	18.6 · max_connections 500 (497 usable) · 49 in use · archive_mode off — no point-in-time recovery
Database size	450 MB. AuditLog is 279 MB of it — 7-year retention, cryptographically chained, cannot be deleted
Disk growth	+100 MB over 7 days. 89% is automated market and round activity, not customers
Connection pool	40 per container. 6 containers = 480 during a rolling deploy, inside the 497 limit
Redis	Connected, subscribed, cross-container. Fail-open behind a circuit breaker; never on the bet path
Real activity today	104 registered · 2–5 betting customers per day · largest market ever: 29 bets from 12 people

Page timing	~510 ms to first byte, <i>identical on every route</i> — the signature of distance, not slow queries
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Two figures in our own internal notes were wrong, both in our favour. The code and documentation state that the database allows 100 simultaneous connections and that Redis is switched off in production. Measured live: the database allows **500**, and Redis is **connected and working across containers**. Several "we cannot scale" conclusions written earlier this year rested on those two stale figures and no longer hold.

Money safety — built, tested, and not to be rebuilt

- **Maintenance mode** pauses new bets and deposits while leaving withdrawals open, so customer funds are never trapped by an operational decision.
- **Payment kill switches** per mobile network, separately for money in and money out.
- **Deposits are settled from a signed re-query, not the incoming message**, exactly once, defended against a tampered amount, and recovered by a 15-second poller and a 5-minute sweep even if the provider's confirmation never arrives.
- **A dead Redis cannot break a bet** — proven by a test that points the platform at a dead port and asserts betting still succeeds.
- **A retry without an idempotency key is refused**, so a retry can never become a second bet.
- **200 simultaneous bets on one market: zero shillings lost or created.**
- **The Up & Down safety net is independent of the game's own off switch** — turning the game off must not disable the thing that returns money already staked in it.
- **Backups are verified by actually restoring them** nightly into a scratch database and re-checking the money totals and the audit chain — not by confirming a file was written.

Recovery position

Question	Today	After the work
Data loss window	Up to 24 hours — one backup a night, nothing between	Minutes
Time to restore	Never measured. Rehearsed once, July, against a database a tenth of today's size	Measured and recorded
Written procedure	None	One-page recovery runbook
Would we be told?	Errors yes. Backup failure no.	Alerts that arrive unprompted
Trade while degraded?	Yes — maintenance mode works today	Unchanged, and written into the procedure

Order of work, and how each is proved

#	Step	Proof it is done
1	Turn on Selcom SMS	Register on a real handset; read the code off the phone
2	Enrol and enforce admin two-factor	Full sign-in with a real code before removing the override

3	Health check, deploy overlap, graceful shutdown	Break the database deliberately and watch the check go red
4	Enlarge the database disk	New size visible; site stays up throughout
5	Cache the busy screens	Re-measure page timing and query rate before and after
6	Move to Amsterdam	Re-run the timing test against today's 510 ms baseline
7	Batch large-market payouts	Settle a test market with over 1,000 winners and time it
8	Turn on point-in-time recovery	Perform a real restore into a scratch database
9	Alerts that arrive by themselves	Break the backup on purpose; confirm the alert lands
10	Rehearse recovery; write the runbook	Record the actual wall-clock recovery time

Step 3 is already built and proved on branch `launch-1k-readiness`: the status page now returns "not ready" when the database is unreachable, tested both ways — red with a dead database, green without one. It is deliberately not yet on the live branch.

Prepared 3 September 2026 against production commit `78c3876e`. Sources: Railway billing API, Railway metrics API, the production PostgreSQL database, the platform's own AI spend ledger, and timed requests to `50pick.tz`. The only unmeasured figure is the Twelve Data subscription, billed outside Railway. Rebuild with `node docs/runbooks/mkpdf.mjs ../launch-1k.html ../50pick-launch-1k.pdf`.